

School of Computer & Systems Sciences
Jawaharlal Nehru University
CS-429, Cyber Security
MCA III – Semester
Mid Semester Examination, 2024

Time: 90 minutes

Max. Marks: 30

Note: Q1 is compulsory. Attempt any five questions from Q2 to Q7. Each Question of 5 marks

- Q1. (a) What is malware.
(b) Give a concrete example where integrity is more important than confidentiality.
(c) Give a concrete example where confidentiality is more important than integrity.
(d) A loss of _____ is the disruption of access to or use of information or an information system or an information system.
(e) Define Kerckhoffs' Principle in the context of cryptography
- Q2. A simple substitution (shift by n) is used and the cipher text CSYEVIXIVQMREXI. Find the key.
- Q3. Decrypt the given ciphertext WWVA using the Hill cipher with a key DCDF.
- Q4. Encrypt the plain text WATERMELON using Affine cipher with the key $a=5$ and $b=8$.
- Q5. Encrypting the message " WE ARE DISCOVERED SAVE YOURSELF " using playfair cipher. The key of the cipher is MONARCHY.
- Q6. Alice uses the same key k to encrypt two messages m_1 and m_2 to get ciphertexts $c_i = E(k, m_i) = k \oplus m_i$ using One Time Pad. Eve later manages to learn the message m_2 in addition to both ciphertexts c_1 and c_2 . Show how Eve can reconstruct m_1 with the available information.
- Q7 What will be the plain text corresponding to cypher text "ROFUTKCAATAT" which obtained using Double transposition cipher with permutation of rows and columns is done according to the specified permutations (1, 2, 3) to (3, 2, 1) and (1, 2, 3, 4) to (4, 2, 1, 3) respectively.